Student Equipment borrowing and return management system

Equipment borrow

Names: Hatunguramye Elichadai

Id:27004

Course: Web Design

AUGUST 5, 2025

Contents of content

1. Project Introduction 1
2. Problem Statement 1
3. System Requirements 2
4. System Design 2
5. Implementation 4
6. Database Design 4
7. Testing 5
8. Challenges Faced 6
9. Conclusion 6
10. Screenshots 6
Overview: ..... **Error! Bookmark not defined.**

1. Project Introduction

Title: Student Equipment Borrowing and Return Management System

Case Study: Built for a university or school to manage equipment borrowed by students (e.g., laptops, projectors, books).

Purpose: The system replaces manual paper-based tracking of borrowed equipment with a reliable online solution. It helps prevent loss, ensures accountability, and improves recordkeeping.

Technologies Used:
Frontend: HTML5, CSS3, JavaScript
Backend: PHP 8, MySQL
Tools: VS Code, XAMPP (Apache + MySQL)

2. Problem Statement

Problem: Many educational institutions face difficulties managing student equipment borrowing manually. Equipment can get lost, double-booked, or records misplaced.

Users:
Students: Borrow and return items

Admin: View, approve, and track all equipment activity

Features that solve the problem:
- Digital forms for borrow and return
- Status tracking (pending, approved, returned)
- Admin panel for review and update
- Availability tracking to avoid conflicts

3. System Requirements

Software:
- Web browser (Chrome, Firefox)
- Visual Studio Code (or any IDE)
- XAMPP (Apache + MySQL)

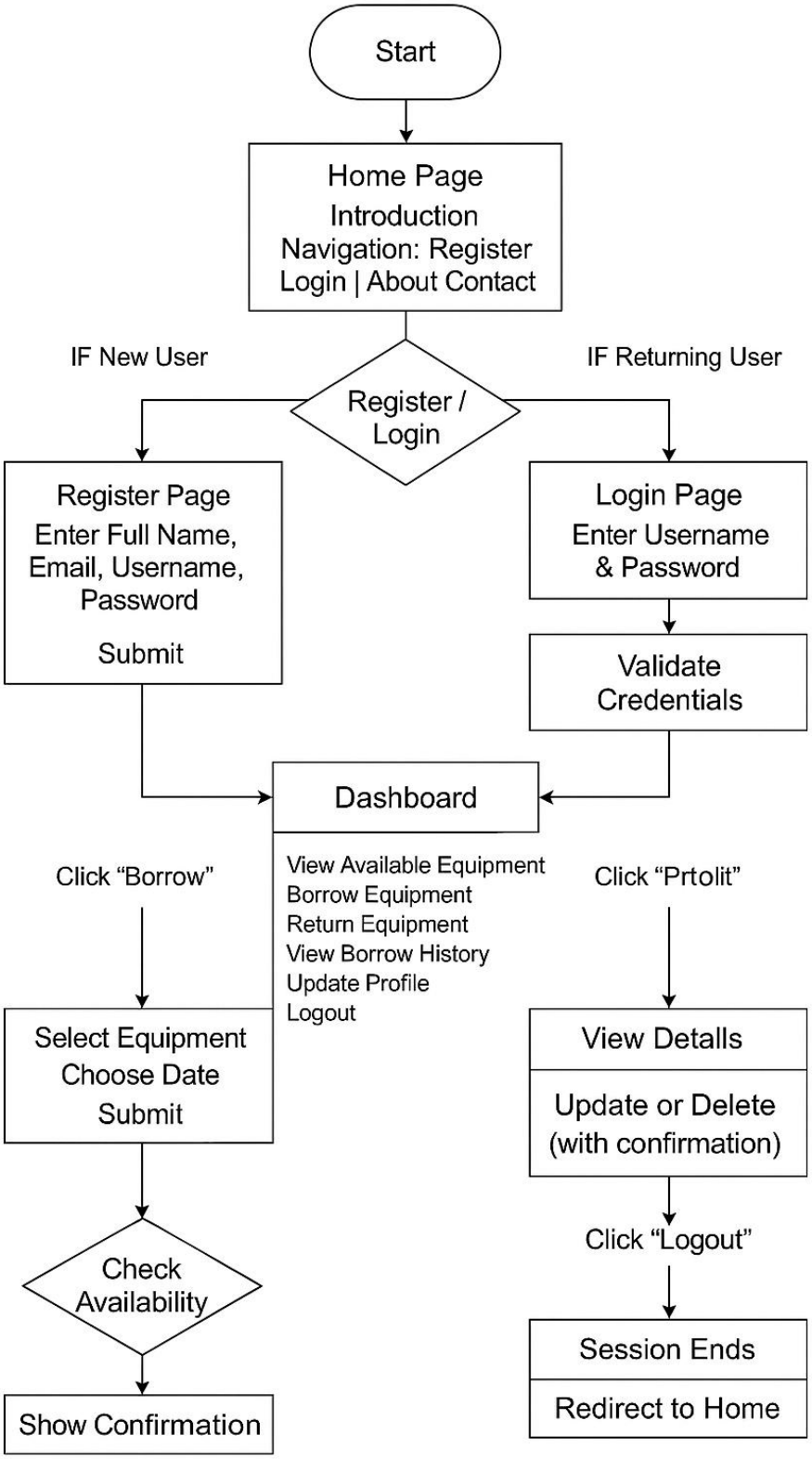
(Optional) Hardware:
- Windows PC or laptop
- 4GB+ RAM recommended

4. System Design

Navigation Flow:
Student → Register/Login → Dashboard → Request Item → Return Item
Admin → Login → View Requests → Mark as Returned

Pages:
- Home (index.html)
- Register (register.html)
- Login (login.html)
- Dashboard (dashboard.php)
- Borrow form (borrow.php)
- Return form (return.php)
- Admin Panel (admin.php)

Flowchart diagram:


Start
Home Page
Introduction
Navigation: Register
Login | About Contact
IF New User
IF Returning User
Register / Login
Register Page
Enter Full Name, Email, Username, Password
Submit
Login Page
Enter Username & Password
Validate Credentials
Dashboard
View Available Equipment
Borrow Equipment
Return Equipment
View Borrow History
Update Profile
Logout
Click "Borrow"
Click "Prtolit"
Select Equipment
Choose Date
Submit
View Detalls
Update or Delete (with confirmation)
Check Availability
Click "Logout"
Show Confirmation
Session Ends
Redirect to Home

5. Implementation

HTML: Structure of forms, dashboard, and admin panel

CSS: Basic styling and table formatting (styles.css)

JavaScript:
- Validates registration/login form fields
- Alerts for incorrect input

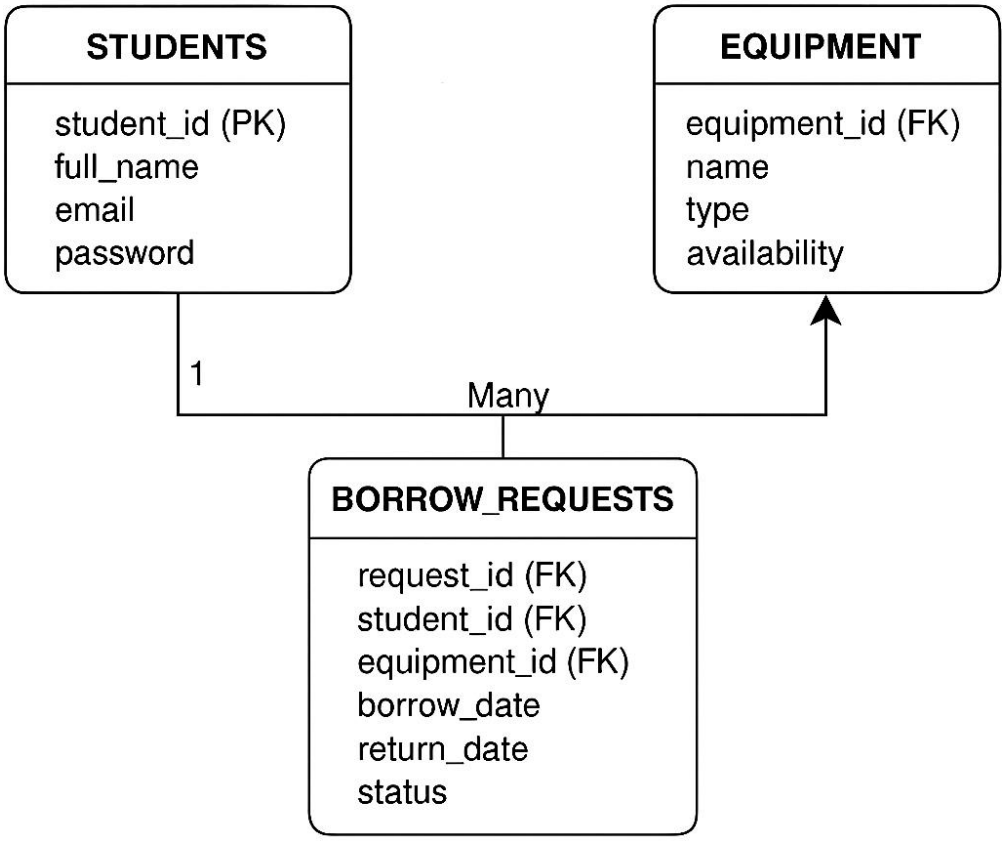
PHP:
- Processes login, borrow, and return
- Uses POST to handle form data securely

MySQL:
- Queries to insert and update records
- Joins to fetch student and equipment names in admin panel

6. Database Design

Tables:
- students(id, name, email, password)
- equipment(id, name, type, availability)
- borrow_requests(id, student_id, equipment_id, date_borrowed, date_returned, status)

ER Diagram:



- One-to-Many from students to borrow_requests
- One-to-Many from equipment to borrow_requests

Relationships:
- borrow_requests.student_id → students.id
- borrow_requests.equipment_id → equipment.id

7. Testing

Methods:
- Manual testing of all forms (registration, login, borrow, return)
- Validation of empty fields and correct email format
- Tested status updates in admin panel

Bug fixing:
- Fixed PHP form errors
- Solved database foreign key constraint errors

- Resolved availability toggle logic

8. Challenges Faced

Challenge 1: Handling foreign key constraints in MySQL
Solution: Created parent tables first and used correct data types

Challenge 2: Maintaining session data for login/logout
Solution: Used PHP sessions with session_start()

Challenge 3: Updating availability after return
Solution: Wrote UPDATE query to mark item as available

9. Conclusion

Summary:

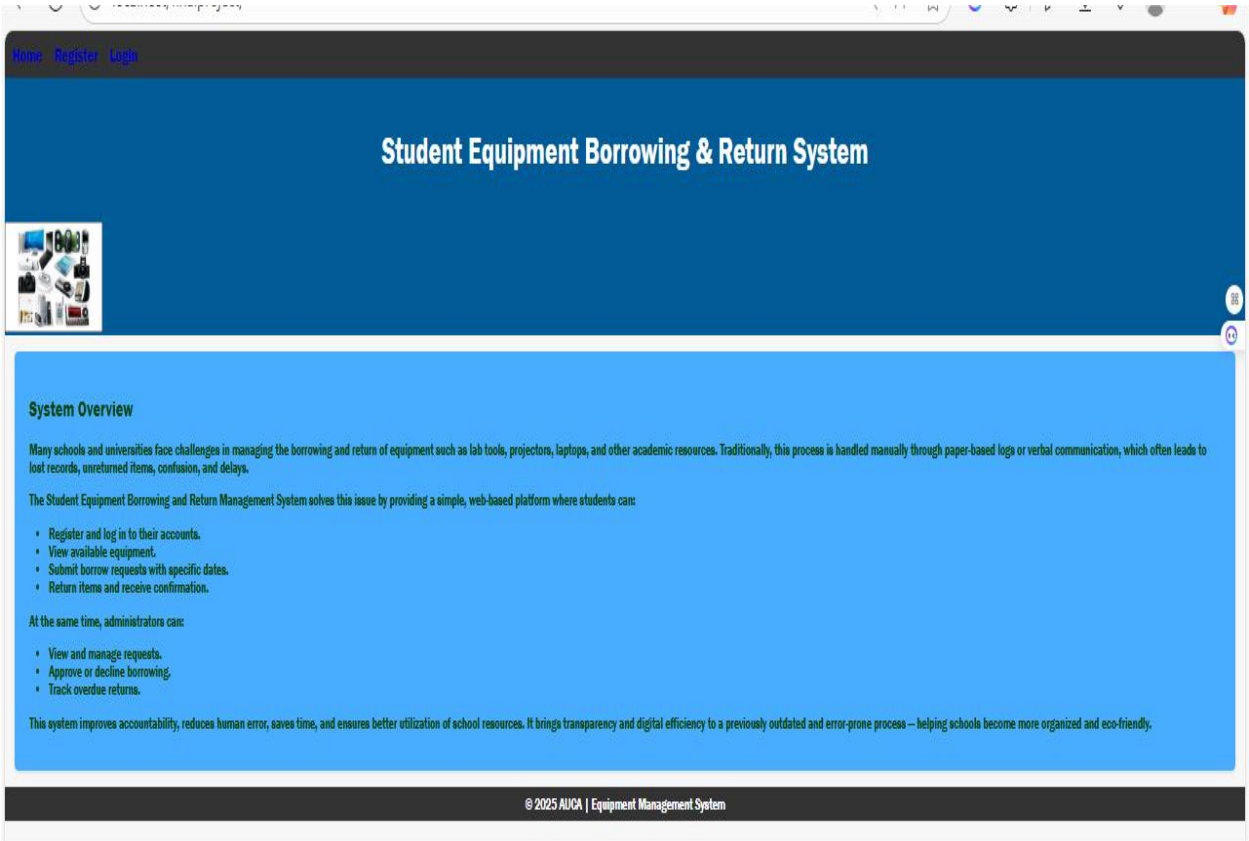
The Student Equipment Borrowing and Return Management System aims to enhance the efficiency of equipment management in educational institutions. By leveraging modern web technologies, the system provides a seamless experience for both students and administrators, reducing errors and improving overall satisfaction. This document serves as a foundational overview of the project, outlining its purpose, features, and design.

Future Improvements:
- Add email notification for approvals
- Enable file uploads for ID verification
- Improve UI with responsive design

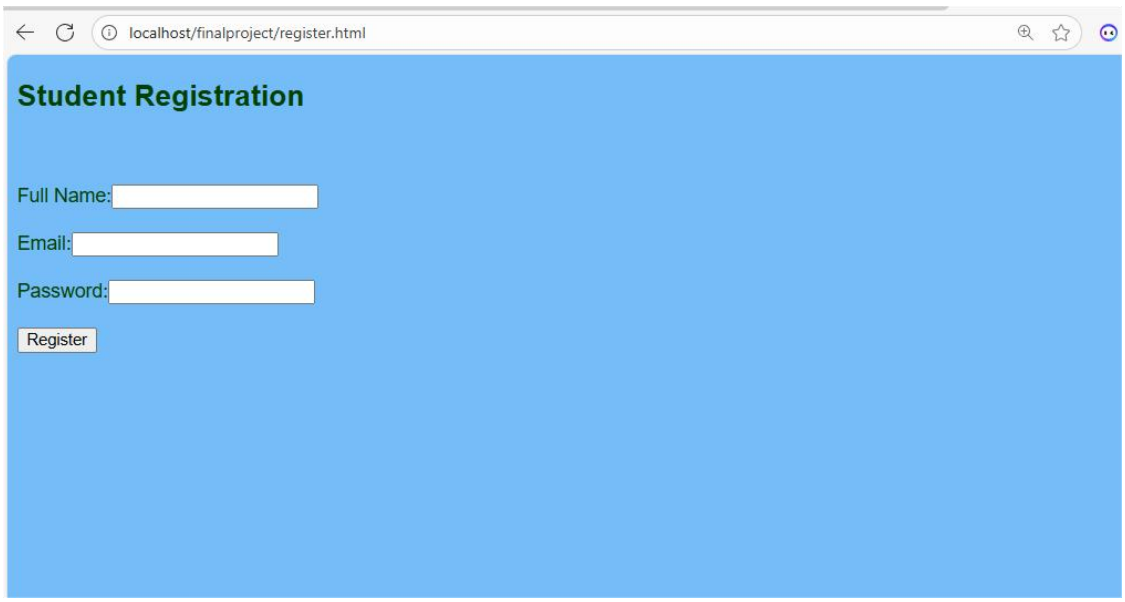
10. Screenshots

Screenshots of the system will be included here to provide visual representation of the user interface and functionalities.

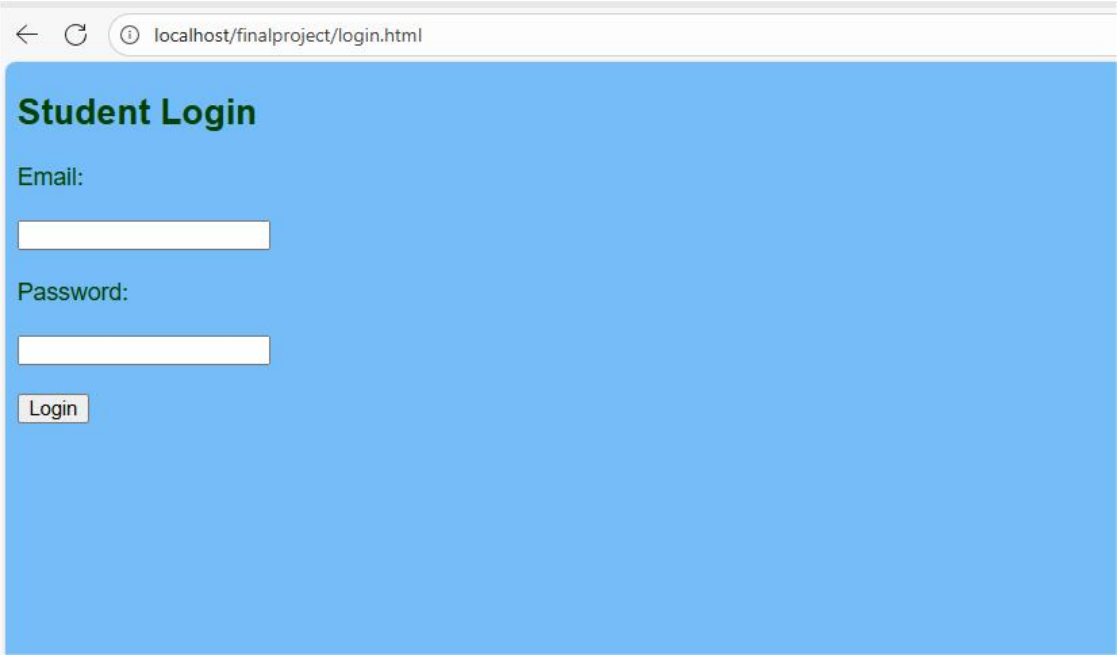
1. # Home Page:



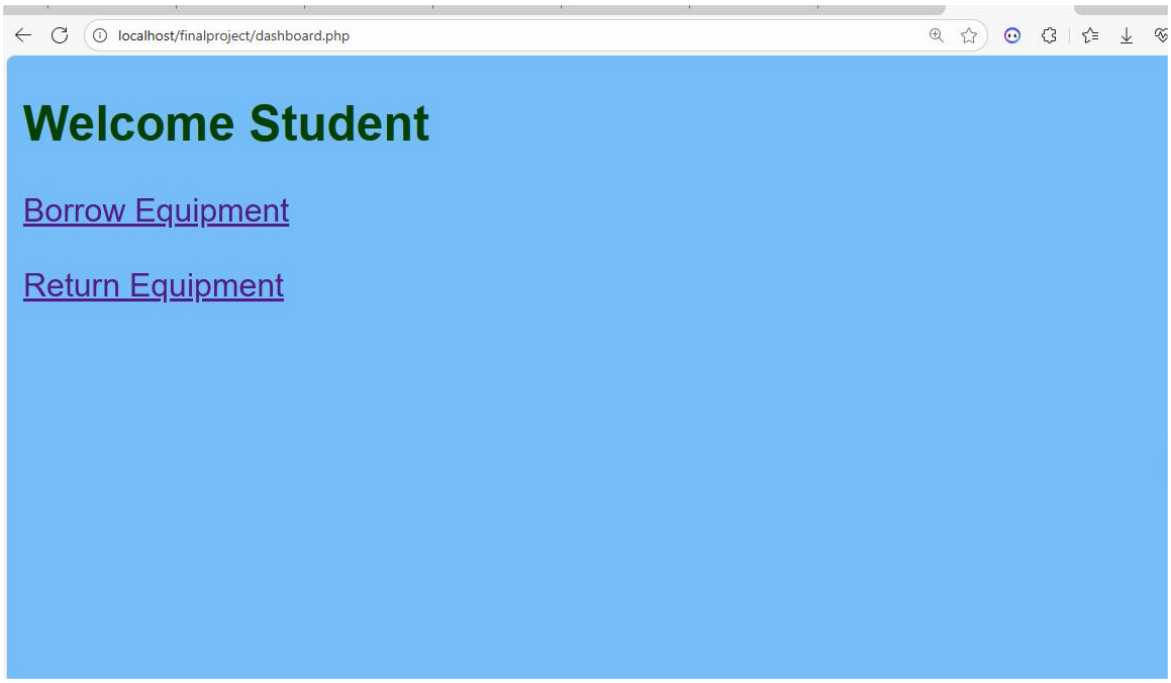
2. # Registration Page:



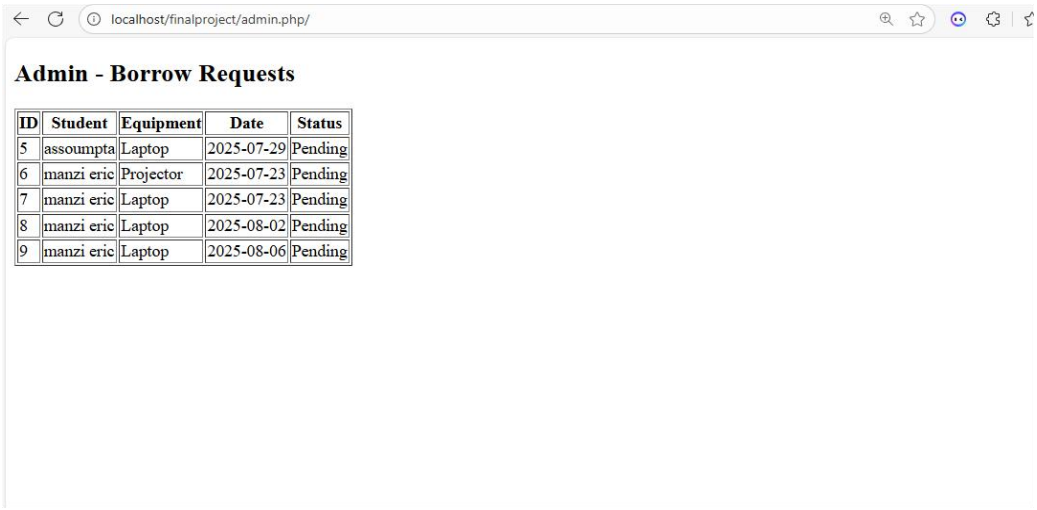
3. # Login page:



4. # Dashboard:

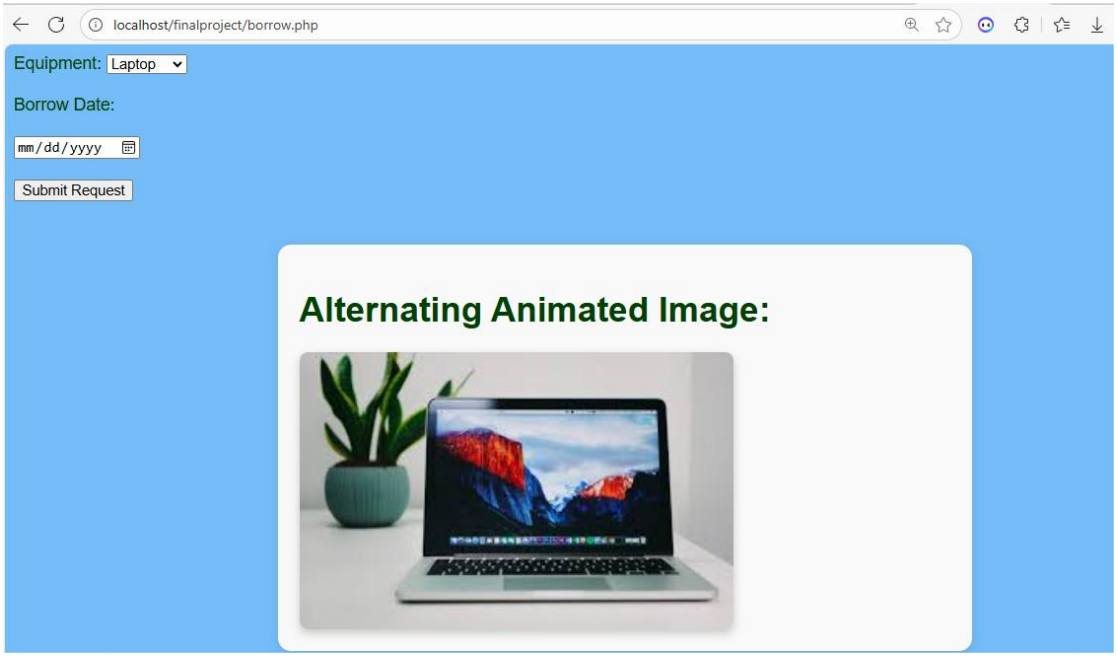


5. # Admin Panel:

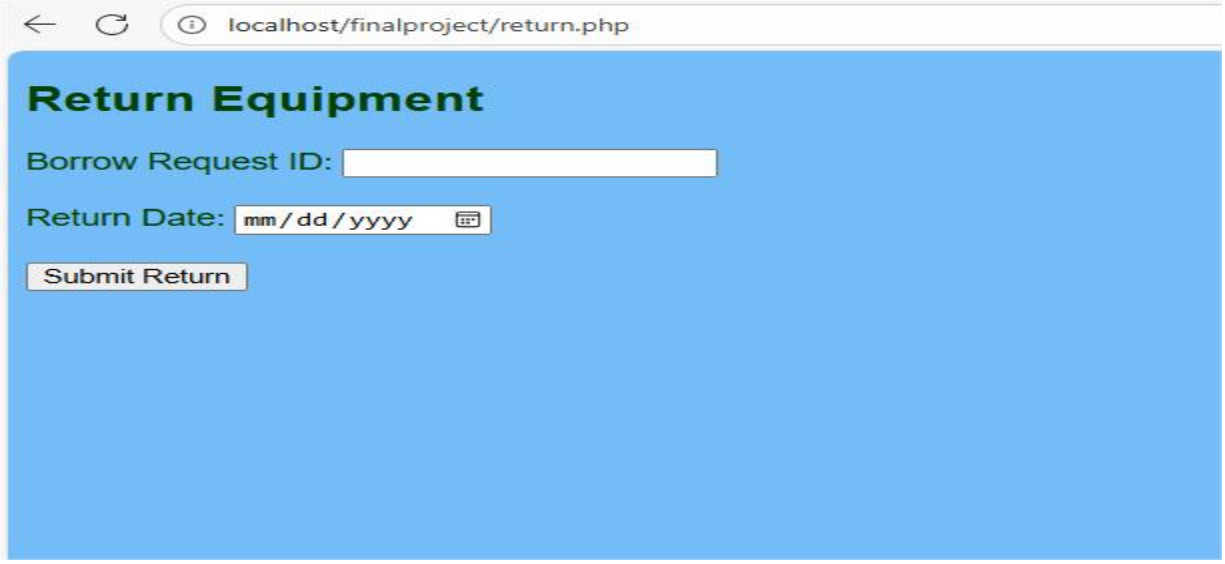
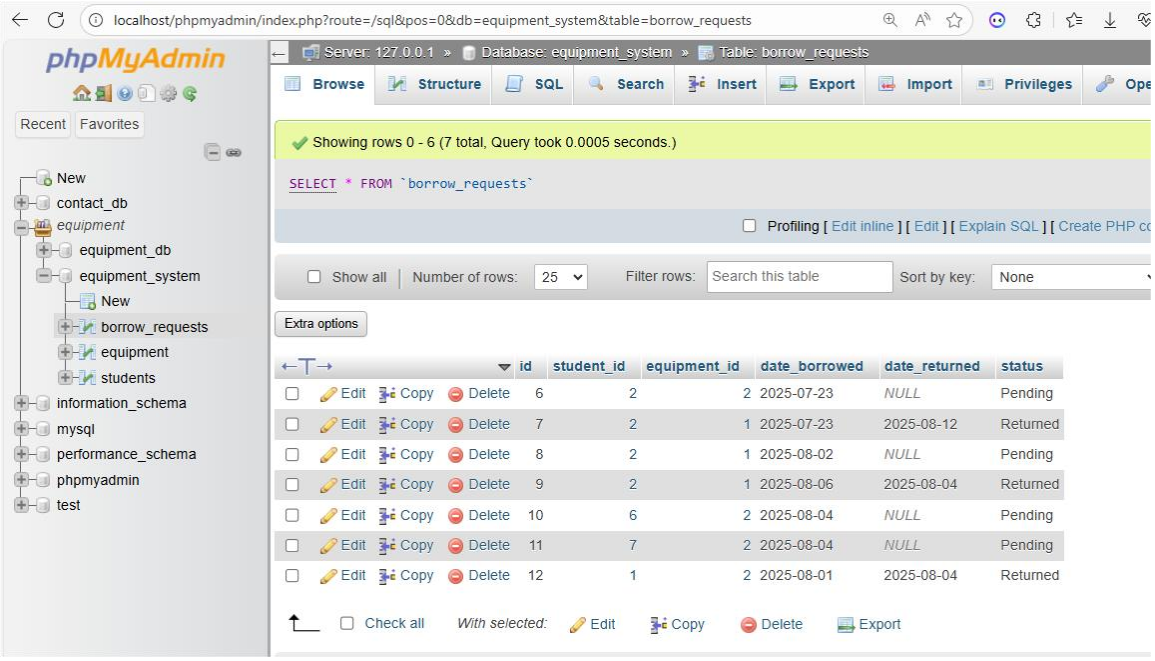


ID	Student	Equipment	Date	Status
5	assoumpta	Laptop	2025-07-29	Pending
6	manzi eric	Projector	2025-07-23	Pending
7	manzi eric	Laptop	2025-07-23	Pending
8	manzi eric	Laptop	2025-08-02	Pending
9	manzi eric	Laptop	2025-08-06	Pending

6. # Borrow Form:



7. # Return Form:



8. Database (phpMyAdmin view of tables)